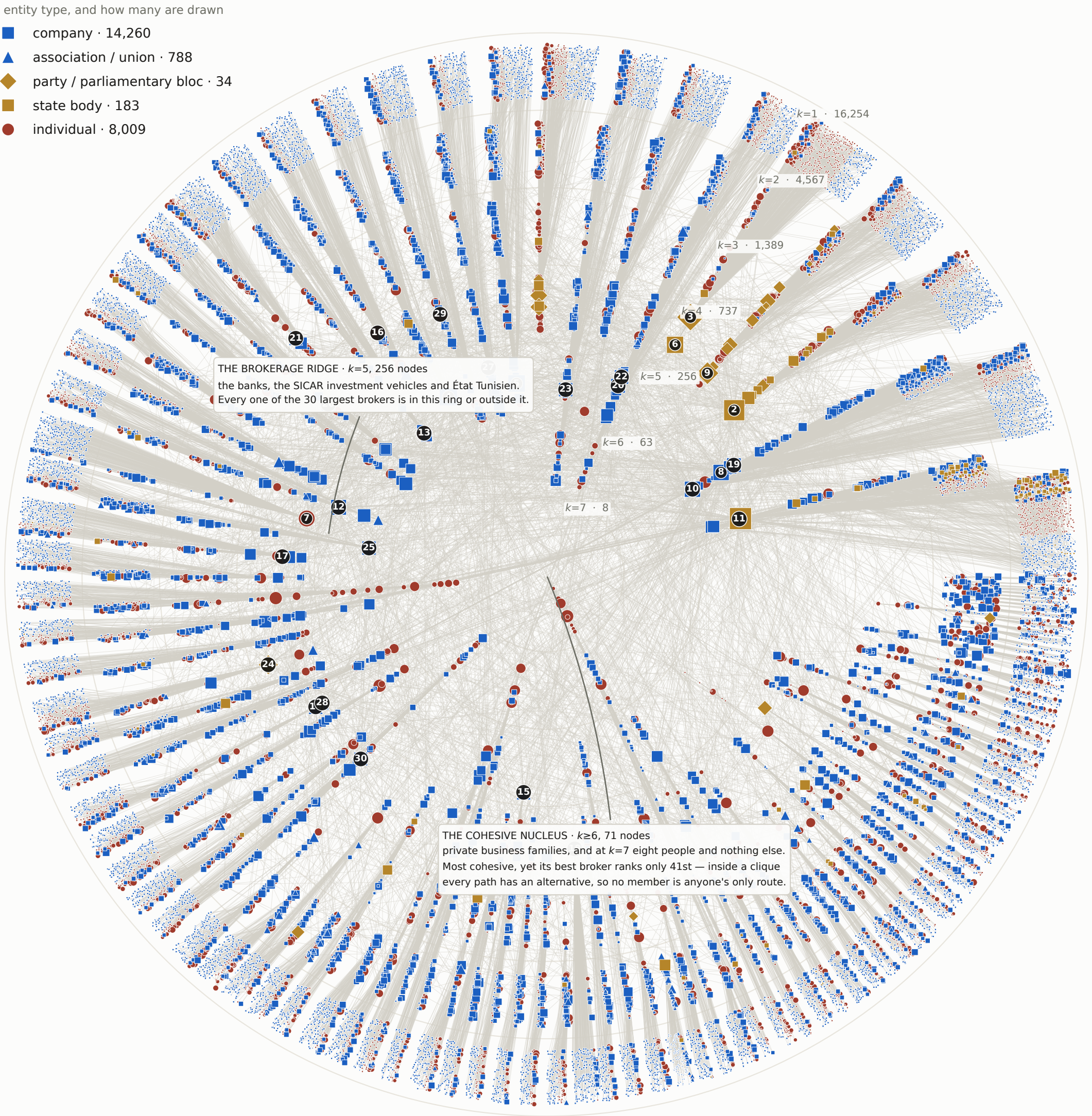


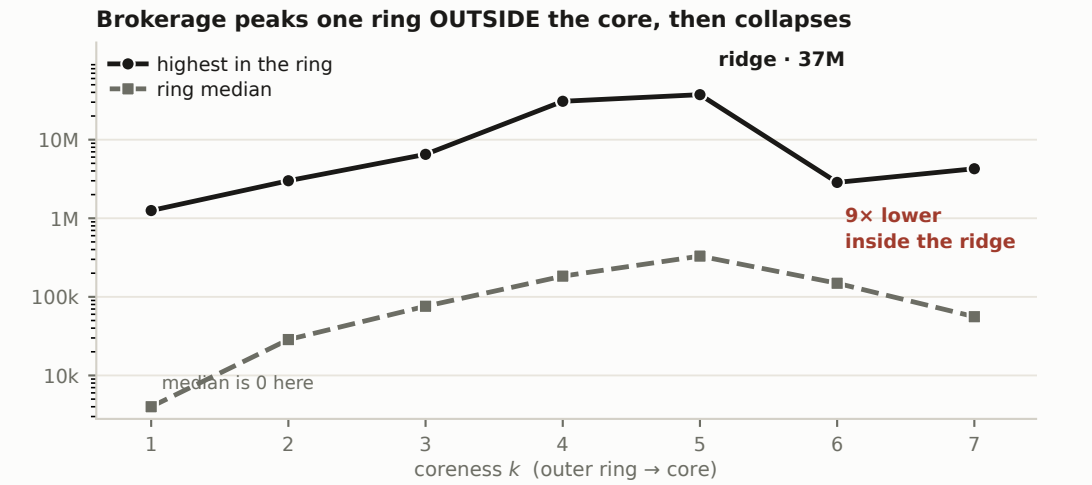
The Tunisian elite network has two centres, and they are not the same people

23,274 entities and 31,457 ties — the largest connected component of the six-category graph. Node area $\propto \sqrt{\text{betweenness}}$; radius is coreness, one equal-width ring per level; angle is community (Louvain, 107 communities, modularity 0.899), so each group reads as a radial wedge. Sizing by brokerage on a radius set by cohesion lets the two disagree — and they do: the biggest nodes are NOT in the centre. Brokerage peaks at $k=5$ (256 nodes, 68% companies: the banks, the SICARs, État Tunisien) and collapses inside it. The $k \geq 6$ nucleus is 71 nodes of private business family — no state body, no party, no association — and its best broker ranks only 41st. Cohesion and brokerage are different properties, held here by different entities.



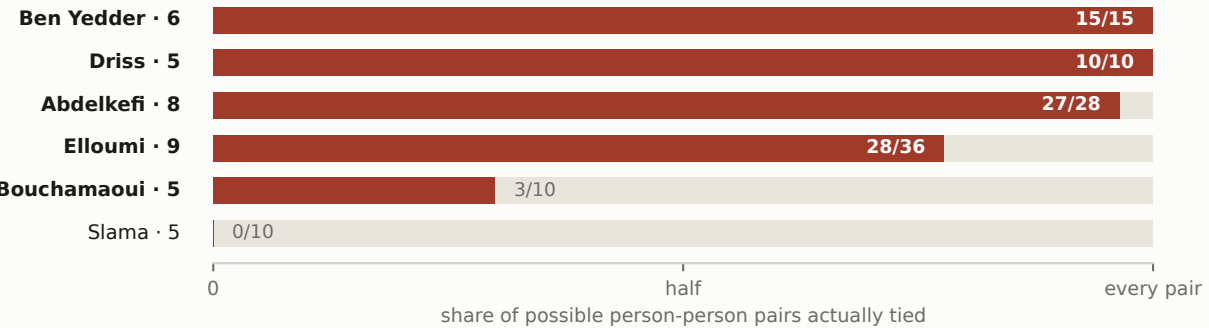
The brokers, numbered on the map betweenness in millions, and the ring they sit in — every one of the 30 is at $k=3-5$, none in the $k \geq 6$ nucleus

1.	Etat Tunisien	37.4M	k=5
2.	Présidence RÉPUBLIQUE	30.8M	k=4
3.	RCD central cttee	22.9M	k=4
4.	Biat	16.5M	k=5
5.	Atd Sicar	13.9M	k=5
6.	Min. ECONOMIE NATIONALE	13.2M	k=4
7.	Mohamed Trabelsi	12.2M	k=4
8.	Sim Sicar	10.7M	k=5
9.	Mouvement Nidaa Tounes	10.5M	k=4
10.	Sicar Invest	10.3M	k=5
11.	Sté TUNISIENNE DE BANQUE	10.0M	k=5
12.	Banque Nationale Agricole Bna	9.4M	k=5
13.	Faycel Derbel	9.1M	k=5
14.	Chambre Tuniso-Francaise De C...	8.1M	k=4
15.	Union Bancaire Pour Le Commer...	7.9M	k=5
16.	Moncef Sellami	7.2M	k=4
17.	Mohamed Ammar	7.1M	k=4
18.	Mohamed Khalil	7.0M	k=4
19.	Stb Sicar	7.0M	k=5
20.	Amen Bank	6.8M	k=5
21.	Associations, Partis, Syndica...	6.5M	k=3
22.	Ennakl Automobiles Sa	6.4M	k=5
23.	International Sicar	6.3M	k=5
24.	Nidaa Tounes Partys Extended ...	6.2M	k=4
25.	Sicar Amen	6.1M	k=5
26.	Sté M	6.0M	k=4
27.	Abdelwaheb Ben-Ayed	5.6M	k=5
28.	Groupe Dimpulsion Economique ...	5.5M	k=4
29.	Abdelkader Hamrouni	5.4M	k=4
30.	Sté DE PROMOTION IMMOBILIERE	5.3M	k=4



The nucleus is cliques — but only where kinship is evidenced

name · members, bar = share of pairs tied. Slama shares only a surname: no kinship tie between any of its 5, which reach $k \geq 6$ on company boards alone.



Betweenness is exactly ZERO for 14,250 of 23,274 nodes (61%) — they lie on no shortest path between any other pair, so they broker nothing. Area is $\propto \sqrt{\text{betweenness}}$ above a visible floor; strict proportionality would erase three nodes in five. Peak betweenness by ring: $k=1$ 1.3M, $k=2$ 3.0M, $k=3$ 6.5M, $k=4$ 30.8M, $k=5$ 37.4M, $k=6$ 2.9M, $k=7$ 4.3M. Coreness and Louvain communities are computed on the undirected simple graph; 267 of the 31,457 ties drawn fall inside the $k \geq 6$ nucleus. Surname blocks in the nucleus are reported one by one rather than pooled, because pooling hides a real distinction: a shared surname is not evidence of a family. 4 of the 6 blocks are near-complete kinship cliques; Slama has none at all. 3 organisation nodes with no label at all are dropped: one carried 295 ties and the HIGHEST betweenness in the graph (76.7M, ahead of État Tunisien), and a node that is not an entity cannot broker anything. Short labels are kept— GAT, MAC and PAF are real firms. Source: Journal Officiel de la République Tunisienne 1957–2026 (jort.tn), the Registre National des Entreprises, and a 13,630-name elite roster. Author's extraction; `scripts/figure\_core\_periphery.py`.